

## Highlights

**What does a released average-subject brain encoder actually predict? A noise ceiling, a parcel-level check, and three silent failures**

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- Leave-one-subject-out ceiling for Friends s01e01a:  $r = 0.1517$ , 1000 parcels
- Released TRIBE v2 average-subject checkpoint:  $r = +0.0283$  over the first 120 s
- Weakly positive,  $p = 0.048$ ; the audit's negative sign is not reproduced here
- Three silent analysis defects changed reported values without raising errors
- Select records by name, fix orientation by atlas size, align clocks in seconds

# What does a released average-subject brain encoder actually predict? A noise ceiling, a parcel-level check, and three silent failures

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## Abstract

*Background:* Released brain-encoding checkpoints are increasingly used as instruments, their predictions read as a proxy for cortical response. A release does not certify that the checkpoint, in the configuration its loader selects, predicts held-out cortex. A self-published audit reported that the released TRIBE v2 checkpoint, in the average-subject configuration its loader forces, is anti-correlated with cortex at vertex level ( $r = -0.0145$  against an inter-subject ceiling of  $+0.0508$ ).

*New method:* A validation protocol on public Algonauts 2025 (CNeuro-Mod) data that runs on CPU from a saved prediction: a leave-one-subject-out noise ceiling measured on the same scans as the encoder, projection of predictions into 1000 Schaefer parcels, alignment of prediction and recording on one clock in seconds, and a circular-shift null.

*Results:* The ceiling for *Friends* episode s01e01a is 0.1517 (95% CI [0.1484, 0.1551]; four subjects, 592 timepoints). Over the first 120 s (80 scans) the checkpoint gives a group mean  $r = +0.0283$  against a same-scan ceiling of 0.0959,  $p = 0.048$  against 61 circular shifts, with three of four subjects positive and the mean positive for response delays of 3 to 7 s.

*Comparison with existing methods:* Index-by-index comparison of a 2 Hz prediction with a 1.49 s recording had produced a plausible, withdrawn  $r = +0.0344$ . Parcel-level values are not directly comparable with the audit's vertex-level values.

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*Conclusions:* On this window the checkpoint is weakly positive and short of decisive. The result does not reproduce the audit’s negative sign; it does not refute it. Three silent defects that changed reported numbers without an error are documented with safeguards.

*Keywords:* encoding models, fMRI, noise ceiling, naturalistic stimuli, temporal alignment, model validation

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## 1. Introduction

A released encoding model invites a particular kind of use: treat the checkpoint as an instrument, run it on stimuli its authors never considered, and read its predictions as a stand-in for measured cortical response. The appeal is obvious, since the alternative is scanning people. The risk is equally obvious once stated, which is that a technical report describing an architecture is not a warranty about the weights released under its name, and that a checkpoint may be released in a configuration whose predictive validity nobody has established.

TRIBE, a trimodal encoder combining pretrained text, audio and video representations, won the Algonauts 2025 challenge [1]. Its successor, TRIBE v2, is released as a checkpoint with a loader that predicts responses on the fsaverage5 cortical surface [2]. The loader in the public repository sets the model to its average-subject configuration at load time, so a downstream user who follows the documented path receives a single group-level prediction.

For this configuration the risk is not hypothetical. A commercial laboratory has published an audit of the released checkpoint in its average-subject configuration [3], reporting vertex-level  $r = -0.0145$  against a measured inter-subject ceiling of  $+0.0508$ : anti-correlated with recorded cortex, across four subjects and all seven Yeo networks [4]. Taken at face value, that would mean downstream studies using this checkpoint as an instrument read a signal with no established relationship to cortex.

The audit should not be accepted uncritically. It is self-published by a company selling the per-subject alternative, it uses four subjects and one stimulus, and it reports magnitudes near 0.015 in either direction, so it is a claim about sign rather than about effect size. Neither should it be dismissed. It is specific, it is falsifiable, and it concerns a configuration that a released loader selects by default.

This paper asks the audit’s question on public data, at parcel level, with a protocol that can be rerun on CPU from a saved prediction. It reports a model-free noise ceiling, a prediction pass over the first 120 s of one episode, and three silent defects met along the way, each of which produced a plausible number without raising an error.

## 2. Materials and methods

### 2.1. Data

We use the publicly released Algonauts 2025 responses [5], recorded by the Courtois NeuroMod project (CNeuroMod) [6]: four subjects (sub-01, sub-02, sub-03 and sub-05) watching the same episodes of the television series *Friends*, sampled every 1.49 s and distributed as time series for the 1000 parcels of the Schaefer atlas [7] (7-network ordering) in MNI152NLin2009cAsym space. All analyses use episode `s01e01a`. Each subject’s recording is truncated to the shortest across subjects, giving 592 timepoints.

### 2.2. Noise ceiling

An encoder’s predictions are uninterpretable without knowing how much stimulus-driven signal is shared across subjects. We estimate it with a leave-one-subject-out procedure: each subject’s parcel time series is correlated, over time and per parcel, with the mean of the remaining three subjects. This is an inter-subject correlation [8], computed from the recordings alone with no model involved. Because the mean of three subjects is itself noisy, the estimate is conservative: a noiseless model of the shared response could exceed it [9, 10]. It is used here as a reference scale, not as a hard upper bound.

Undefined correlations stay undefined: a parcel with no variance over time has no Pearson correlation, and substituting zero would pull the mean toward zero with invented values. Parcel values are averaged within each subject, and the group value is the mean over subjects. Confidence intervals are percentile bootstrap over subjects (10,000 resamples, fixed seed), so a reported interval reproduces exactly.

### 2.3. Checkpoint and prediction

The released TRIBE v2 checkpoint [2] is loaded through its own loader, which forces the average-subject configuration. The checkpoint emits one

prediction per 0.5 s of stimulus (its feature rate is 2 Hz) for 20,484 fsaverage5 vertices. The encoder pass is bound by video feature extraction, at about 17 minutes per 60 s of video on the GPU available (an NVIDIA Tesla P100 in a free-tier session). A whole-episode pass exceeded the 12-hour session limit before producing output, so the first 120 s of s01e01a were predicted, giving 241 prediction samples. The saved prediction is the input to every analysis below, all of which run on CPU.

Predictions are projected from fsaverage5 vertices into the 1000 Schaefer parcels by averaging the vertices carrying each parcel label in the surface version of the atlas [7]; all 1000 parcels receive vertices. The prediction is projected down rather than the recordings upsampled, which would invent within-parcel structure the data do not have.

#### 2.4. Temporal alignment

The checkpoint was trained against BOLD shifted back by 5 s (the offset in its training configuration), so the prediction for stimulus time  $t$  estimates the scan recorded at  $t + 5$  s. For each recorded scan  $j$ , the stimulus time it responds to is  $1.49j - 5$  s. The prediction is linearly interpolated at that time for every scan whose stimulus time lies inside the predicted window; scans outside it are dropped, not extrapolated. For the 120 s window this leaves 80 scans, recording indices 4 to 83. The noise ceiling is then recomputed on exactly these scans, so encoder and ceiling share one denominator.

The 5 s delay was taken from the checkpoint’s training configuration and fixed before any correlation was computed. Delays of 3, 4, 6 and 7 s are reported as sensitivity and are not selected from.

#### 2.5. Statistical inference

The encoder score for each subject is the mean over parcels of the Pearson correlation, over time, between the aligned prediction and that subject’s recording. The sign is reported as measured; nothing is flipped or absolute-valued.

A bootstrap over four subjects has only 35 distinct resamples, so the encoder’s bootstrap interval is reported but not relied on. The test is a circular-shift null [8]: the aligned prediction is rotated against the recordings by every shift from 10 to 70 scans (61 shifts), which preserves each series’ autocorrelation and breaks only their alignment, and the group mean correlation is recomputed at each shift. The one-sided  $p$  is  $(b + 1)/(m + 1)$ , where  $b$  is the number of the  $m$  shifts reaching the observed value [11].

## 2.6. Software

Analyses use Python with NumPy and h5py. The noise ceiling, prediction pass and null are separate scripts (see Data and code availability); each episode is matched by name and each array’s orientation is asserted against the atlas size of 1000 parcels.

## 3. Results

### 3.1. Noise ceiling on the full episode

For **s01e01a**, over four subjects, 1000 of 1000 parcels defined and 592 timepoints,

$$r_{\text{ceiling}} = 0.1517, \quad 95\% \text{ CI } [0.1484, 0.1551]. \quad (1)$$

Per-subject values are 0.1553, 0.1548, 0.1497 and 0.1470.

This is larger than the audit’s +0.0508, and the difference is not a disagreement. Averaging vertices within a parcel cancels independent noise and generally raises correlations, and the two ceilings were measured on different stimuli and data releases, so a parcel-level ceiling here and a vertex-level ceiling there are not comparable numbers.

### 3.2. Prediction pass on the first 120 seconds

Table 1 gives the aligned prediction pass.

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| Encoder, group mean $r$              | +0.0283                              |
| Per subject (sub-01, 02, 03, 05)     | −0.0048, +0.0316, +0.0541, +0.0322   |
| Encoder bootstrap CI (not relied on) | [+0.0045, +0.0485]                   |
| Noise ceiling on the same 80 scans   | 0.0959, CI [0.0885, 0.1025]          |
| Shift null (61 shifts)               | mean −0.0048, sd 0.0191, max +0.0351 |
| One-sided $p$                        | 0.048                                |

Table 1: Aligned prediction pass over the first 120 s of **s01e01a**, Schaefer 1000 parcels, 80 scans (recording indices 4 to 83), response delay 5 s.

Three of four subjects are positive and one is indistinguishable from zero. Two of the 61 shifted alignments reach the observed value, giving  $p = 0.048$ , at the margin of the conventional threshold. Across response delays of 3 to 7 s (80 or 81 scans) the group mean stays positive, from +0.0239 to +0.0377, so the sign does not depend on the delay chosen. The ceiling on these 80 scans is lower than the full-episode 0.1517; it is measured on fewer and earlier scans, and the two are never mixed.

### 3.3. Three silent failures

Three defects in this work changed a reported number without raising an error. Each was identified the same way: replaying the original code path on the same inputs reproduces the original number exactly, which establishes the cause rather than merely fitting it. The values below are withdrawn and are reported only to document the failure.

*The episode was selected by position.* An earlier ceiling of 0.2247, CI [0.2147, 0.2363], described as covering “482 parcels”, is withdrawn. The analysis code carried a placeholder for the stimulus identifier and fell back to the first key in the response file, which holds a different episode (s01e02a) from the one under prediction. Subjects also do not agree on which session an episode is stored under, so positional selection is inconsistent across files as well.

*The array was transposed.* The correlation routine expects arrays shaped as units by timepoints. The recordings are stored as timepoints by parcels, and the orientation guard tested whether the first axis was longer than the second. For a (482, 1000) array that test cannot fire, so every correlation ran across the parcel axis. The “482 parcels” in the withdrawn figure was the timepoint count. Replaying the original path reproduces 0.2247, its interval and the count of 482 exactly.

*Two clocks were compared index by index.* The first completed prediction pass reported an encoder correlation of +0.0344, which is withdrawn. The analysis truncated the 241-sample prediction and the recording to the shorter length and correlated them sample by sample, so prediction  $k$ , at stimulus time  $0.5k$  s, was scored against scan  $k$ , recorded at  $1.49k$  s. By the last sample the two were about four minutes apart. Replaying that path on the saved prediction reproduces +0.0344 and all four per-subject values exactly. A synthetic check shows the size of the error: two series drawn from one signal, one sampled as the prediction is and one as the recording is, correlate at  $r > 0.999$  once aligned and at  $r = -0.03$  when truncated and compared index by index.

## 4. Discussion

### 4.1. Relation to the audit

On this window the released checkpoint is not anti-correlated with the recordings in parcel space. That is the full extent of the claim. The result

does not reproduce the negative sign the audit reports, but it does not refute the audit either. The audit measured vertices on a different stimulus; this work measured parcels on 120 s of one episode. Parcel averaging raises correlations, so a positive parcel-level value is compatible with a vertex-level value near zero, and  $p = 0.048$  from a single window of 80 scans is weak evidence that one further window could overturn. A strict replication requires vertex-level CNeuroMod derivatives and the audit’s stimulus.

#### *4.2. Comparison with existing methods*

The noise ceiling used here is the leave-one-out inter-subject correlation common in naturalistic fMRI [8]. It is conservative relative to estimators that correct for noise in the reference, such as extrapolation from repeated responses [9] or the analytical and split-half ceilings of Lage-Castellanos et al. [10], which rely on repeated presentations or split data. The circular-shift null follows established practice for autocorrelated time series [8], with the  $p$ -value computed so that it cannot be zero [11]. The departure from common practice is the alignment step: equal array lengths are often taken as evidence of equal sampling, and in this work that assumption alone produced a plausible positive correlation from misaligned series.

#### *4.3. Recommendations*

All three defects are generic, so we state three practices. Select records by name, never by position. Decide array orientation against a known constant such as the atlas size, never by comparing axis lengths. Place prediction and recording on one clock, in seconds, before correlating anything, and measure the ceiling on the same scans as the encoder.

#### *4.4. Limitations*

The prediction covers 120 s of one episode, 80 scans, and four subjects; the result is marginal and may not hold over the full episode. Bootstrap intervals over four subjects, including those on the ceilings, have few distinct resamples and understate uncertainty. The prediction is projected into parcels through the surface version of the Schaefer atlas while the recordings use its volumetric version in MNI space; the parcels correspond by label but are not sampled identically. The 5 s response delay comes from the checkpoint’s training configuration and is a single shift rather than a fitted haemodynamic response. The average-subject prediction is one time series compared against every subject, so it cannot capture individual differences by design. Only the checkpoint’s default configuration was tested.

#### 4.5. Future work

The immediate next step is the same aligned pass over the whole of s01e01a and further held-out episodes, which requires GPU time beyond a single free-tier session. Vertex-level comparison against the audit would require CNeuroMod surface derivatives.

### 5. Conclusion

The noise ceiling for the target episode is 0.1517, CI [0.1484, 0.1551], measured from public data and independent of any model. The aligned prediction pass over the first 120 s gives a group mean of +0.0283 against a ceiling of 0.0959 on the same scans, at  $p = 0.048$  against a circular-shift null: positive, weak, and short of decisive. It does not reproduce the negative sign the audit reports, and at parcel level it is not directly comparable with the audit’s vertex-level  $-0.0145$ . Three withdrawn values and their causes are reported because each was silent, each is reproducible by anyone attempting this measurement, and none would have announced itself.

#### Ethics statement

No new data were collected. The study uses publicly released fMRI data from the CNeuroMod project, distributed through the Algonauts 2025 challenge, which were collected with participants’ informed consent by that project and are used under the terms of their release.

#### Data and code availability

Code and derived outputs are public at <https://github.com/brn-mwai/monarch>. The ceiling is produced by `services/inference/scripts/paper3_noise_ceiling.py`; the prediction pass by `services/inference/scripts/paper3_prediction.py`, which accepts a saved prediction so the analysis can be rerun without a GPU; the alignment by `services/inference/app/services/temporal_alignment.py`; and the null and delay sensitivity by `services/inference/scripts/paper3_alignment_checks.py`. The reported (non-withdrawn) values are recorded in `services/inference/data/paper3/` and `services/inference/data/final/paper3/`. The saved 120 s prediction is available from the author on request. The Algonauts 2025 and CNeuroMod data are not redistributed here and are available from their providers under their own terms [5, 6].

## **CRedit authorship contribution statement**

**Brian Mwai:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

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